Vasile Grafu

Engineering Manager | Solutions Architect | Software Architecture, Cloud & Agentic AI

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SUMMARY

21+ years in software development, and every phase of it: shaping the specification, designing the architecture, writing the code, holding the quality, shipping the release, and running it in production. Usually several of those at once.

- Millions of requests served every day by the platform my team runs
- 50 web services in production, distributed across AWS
- 9 years leading teams, two of them running two at once
- 10,000+ business clients on the systems I have built and led
- 21+ years, from real-time game engines to agentic AI systems

SKILLS

AI & Machine Learning: Agentic systems in production — and the theory underneath them. AI-Assisted Software Development, Agentic Architectures & LLM Orchestration, RAG (Retrieval-Augmented Generation), MCP Gateways, MCP Servers & tool integrations, Machine Learning, Neural Networks & Deep Learning Foundations

Cloud, Data & DevOps: Running systems reliably on AWS, from database to pipeline. AWS (EC2, ECS, Fargate, RDS, S3, CloudFormation, CDK), Git, GitHub, GitHub Actions, Docker, CI/CD, End-to-end testing (pytest, Playwright), SQL Server, PostgreSQL, Database Design, Relational Databases

Frontend & Web: Interfaces for the platforms and tools I build. JavaScript, TypeScript, jQuery, ReactJS, Material UI, HTML, CSS, SASS, Astro, Tailwind CSS, HTMX, Server-Rendered Web (ASP.NET MVC + Razor, Flask + Jinja)

Backend & Architecture: Designing and building the services that carry the business. C#, .NET, ASP.NET Core, Entity Framework, LINQ, Python, FastAPI, Flask, SQLAlchemy, Solution & Software Architecture, Framework Design, API Design (REST, SOAP/WCF)

Leadership & Management: Building, growing, and running engineering teams. Engineering Management, Team & Technical Leadership, Hiring & Mentoring, Agile / Scrum, Project Management (Prince2), Strategic Planning

Foundations: The math and fundamentals everything else stands on. Probability, Linear Algebra, Calculus, Statistics, Data Structures & Algorithms, Problem Solving & Analytical Thinking, Software Architecture Principles, Design Patterns & Clean Code

EXPERIENCE

Engineering Manager · nShift (formerly Consignor), Bucharest, Romania

January 2022 – Present

nShift is a global leader in delivery and experience management software, formed in 2021 from the merger of Consignor, Unifaun and Transsmart. Its cloud platform covers the whole parcel journey — checkout delivery options, rule-based carrier selection, label and document generation, tracking, and returns — connecting retailers, 3PLs and manufacturers to a network of over 1,000 carriers across some 190 countries, with integrations into 450+ OMS, ERP and WMS systems.

I lead the Webservices Team — one of the company's core teams — building applications that handle millions of requests per day and support multiple business-critical systems. Most weeks I wear several

hats at once: software architect, software engineer, AI engineer and engineering manager. In practice that means owning the architecture, writing a large share of the code, building the AI systems on top of it, and still setting direction for the team.

- **Software architecture:** design and implementation of scalable, secure, and maintainable systems.
- **Software engineering:** shared components, application structure, data flow, database design, security, and infrastructure.
- **AI engineering:** the agentic systems built on top — LLMs, agent orchestration, RAG, and MCP servers and tool integrations.
- **Solution Architect:** cloud environments on AWS, and the CI/CD pipelines that deliver to them.
- **Test automation:** helped build the end-to-end test framework covering the APIs and the UI, and the tests that run on it.
- **Lead and manage the Webservices Team:** setting clear goals, providing direction, and keeping every member motivated and effective.
- **Establish engineering processes and standards:** consistent development practices that improve collaboration, maintain quality, and accelerate delivery.
- **Provide technical guidance:** solving complex issues alongside developers, reviewing designs, and making critical decisions on challenging topics.
- **Recruit and develop engineers:** active role in hiring, onboarding, and mentoring to build a skilled, cohesive team.
- **Act as Scrum Master:** stand-ups, planning, retrospectives, and effective communication between Product Owners and developers.

TECH C# / .NET, ASP.NET Core · Python, FastAPI, SQLAlchemy · SQL Server, PostgreSQL · JavaScript, TypeScript, ReactJS, Material UI · LLMs, Agents, RAG, MCP · AWS (EC2, ECS, Fargate, RDS, S3, CloudFormation, CDK) · Git, GitHub, GitHub Actions · Docker · pytest, Playwright · AI-assisted engineering

Team Manager · Consignor (formerly Edisoft), Bucharest, Romania

January 2017 – January 2022

Founded in Oslo in 1997, Consignor grew into one of the largest suppliers of transport administration and multi-carrier shipping software in Scandinavia, serving more than 10,000 business customers from offices in Norway, Sweden, Denmark, Finland, Romania and the UK. In 2021 it merged with Unifaun and Transsmart to become nShift.

I led two teams — the Webservices Team and the Integration Team — and wore the hat each situation demanded: Team Manager, Technical Lead, Project Manager (Prince2), or Software Developer. In practice that meant hiring and training the people who joined, owning the critical architecture decisions, running projects across team boundaries — and still writing code myself.

- **Webservices Team:** ran the web services serving the client components of the Consignor business.
- **Integration Team:** brought Consignor functionality to the major e-commerce and checkout platforms — WooCommerce, Magento and Klarna.
- **Software architecture:** the design decisions across the Consignor suite — service boundaries, shared components and the data model.
- **Software engineering:** shared components, application structure and database design across the suite, and the PHP integrations into the e-commerce platforms.
- **Solution Architect:** the AWS environments the platform ran on, from EC2 to Docker containers on ECS and Fargate, and the CI/CD pipelines that deployed to them.

- **Team leadership:** set goals and direction for both teams, keeping two groups with very different missions pulling toward one platform.
- **Technical guidance:** worked through the hard problems alongside the developers.
- **Recruit and develop engineers:** hiring, onboarding, and training the people who joined either team.

TECH C# / .NET, ASP.NET Core · PHP · AWS (EC2, ECS, Fargate, S3, CloudFormation) · Git, Bitbucket, GitLab · Docker · Project Management (Prince2)

Technical Lead · Consignor (formerly Edisoft), Bucharest, Romania

2010 – 2017

I led the Webservices Team, behind the 50 web services that served over 10,000 business clients in the Consignor application suite.

- **Applications at the core of the platform:** built Routing Codes, Address Lookup and Sending Edifacts — all still part of the nShift ecosystem today.
- **Custom Price Calculation Engine:** kept developing the engine I had built the year before, as its use spread across the platform.
- **Software architecture and solution design:** owned the design decisions across the Consignor suite, from service boundaries to data model.
- **Hands-on throughout:** wrote a large share of the code myself, alongside the team rather than above it.
- **Mentoring:** brought on junior and senior developers, and helped hire the people who joined.

TECH C# / .NET, ASP.NET MVC, WCF · SQL Server · JavaScript, jQuery · AWS (EC2, S3) · SVN and Git (Bitbucket, GitLab)

Senior Software Developer · Edisoft, Bucharest, Romania

2009 – 2010

Edisoft was later rebranded as Consignor — the Scandinavian multi-carrier shipping software business that merged with Unifaun and Transsmart in 2021 to become nShift. The name changed; the platform and the team did not.

I designed and built the Custom Price Calculation Engine, and when the manager left the following year I took over the Webservices Team as Technical Lead.

- **Custom Price Calculation Engine:** designed and built it — customers upload the rates they have agreed with their carriers, the engine parses that price sheet, and it calculates the price of each shipment from it.
- **Still in production:** written in 2009 and running today across nShift, where thousands of customers depend on it.
- **Outgrew one developer:** the application became large enough to need its own team, first at Edisoft and later Consignor, building price sheets with customers and training them on it.

TECH C# / .NET, ASP.NET Web Forms, WCF · SQL Server · SVN

Team Leader · TeamNet, Bucharest, Romania

2008 – 2009

TeamNet International, a Romanian technology and systems-integration company delivering software, IT infrastructure, and technology solutions for major public- and private-sector organizations.

The title here covered module ownership rather than people: I took Optimus ERP modules end to end, from specification through testing and deployment. Leading a team came later, at Consignor.

- **Optimus ERP modules:** managed them through every development step, from specification to testing and deployment.
- **Feature planning:** built development plans for new features with the design department.
- **Quality and maintenance:** enhanced existing features, and triaged and resolved every defect raised by the Quality Department.

TECH C# / .NET, WinForms · SQL Server

Game Programmer · Ubisoft, Bucharest, Romania

January 2005 – June 2008

Ubisoft is a French video game publisher founded in 1986 by the Guillemot brothers and now one of the largest in the world, with studios across more than 45 locations behind franchises such as Assassin's Creed, Far Cry, Rainbow Six and Prince of Persia. Ubisoft Bucharest, opened in 1992, was the company's first production studio outside France and grew into one of its largest.

I designed, developed, and optimized gameplay and platform modules on four shipped and in-development titles.

- **Silent Hunter 4:** designed and built the framework managing the evolution of the Crew, Weapons, and Equipment of the U-Boat simulator.
- **King Kong (PC):** Windows Media Center integration, multithreaded rendering, and the security module.
- **Blazing Angels (PC):** Save/Load system and Media Center shell implementation.
- **Tom Clancy's FireHawk:** ERS (Enhanced Reality System) module.

TECH C++, STL · Linear algebra, Calculus, Probability

PROJECTS

Conversational AI Platform

Engineering Manager & Architect · nShift

A large-scale AI assistant built on an agentic architecture: LLMs combined with orchestration layers, RAG, and structured prompt engineering. Domain-specific agents for different departments — each with tailored prompts, tools, and data access — collaborate through an orchestration layer. A continuous ingestion pipeline indexes internal documentation, APIs, and business data into knowledge stores, so responses stay context-aware and grounded in company-specific information.

Per department — agents with their own prompts, tools and data · Continuous — ingestion of docs, APIs and business data · Grounded — answers from company-specific knowledge

TECH LLMs · Agents · RAG · MCP · Orchestration · Python

AI App Builder — Claude Code Skill

Independent R&D · Personal project

An advanced Claude Code skill that builds applications from existing codebases instead of from imagination. It reads codebases of any size — Python, TypeScript, JavaScript, C# — through a structural snapshot built once and queried many times, then works in three modes: generating new code shaped like the code that already exists, migrating an application to another stack, or translating it to another technology while preserving its domain and structure. Every output is proven rather than assumed — contract conformance, entity preservation, and side-by-side behaviour parity — and every decision that shapes the result is put to the user, never guessed.

4 — languages read: Python, TypeScript, JavaScript, C# · 3 — modes: generate, migrate, translate · Proven — contract, entity and behaviour parity checks

TECH Claude Code · Agent Skills · Code Generation · Static Analysis · Python

EDUCATION

Master of Science in Computer Science, Specialization in Project Management — Academy of Economic Studies, Bucharest (2000 – 2002)

Bachelor in Business Administration, Major in Computer Science, Specialization in Computer Science Applied in Business — Academy of Economic Studies, Bucharest (1995 – 2000)